

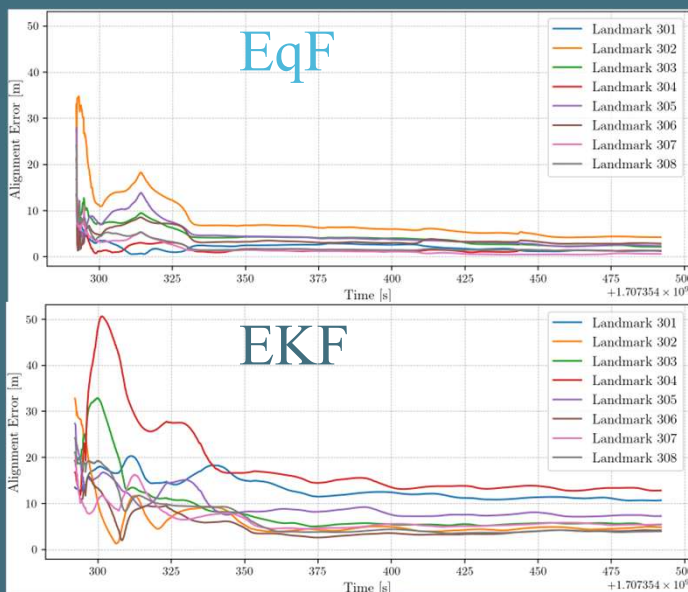
Underwater SLAM system with equivariant (EqF) filter

? The problem

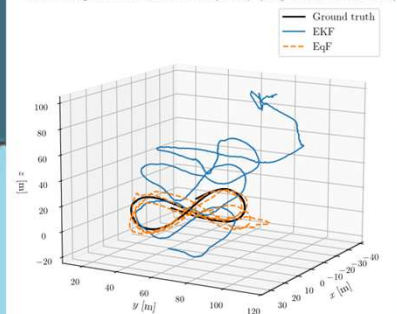
- Commercial Ultra-Short Baseline (USBL) localization systems offers 0.1% to 0.5% accuracy while Long Baseline (LBL) offers 0.05% to 0.2% accuracy. However, it is expensive to survey pre-deployed reference points that is used such systems.
- Alternative SLAM systems usually relies on filters that linearize the kinematics such as Extended Kalman filter (EKF)

✓ Our solution

- Our SLAM system is cheaper than commercial localization method and uses an EqF providing better accuracy and robustness comparing to EKF.
- EqF with Scaled orthogonal transformations SOT(3) works perfect for beacon range measurements because it decouples the bearing information which is highly uncertain and depth information which is very accurate.



EKF vs EqF vs Ground Truth Trajectory (Aligned on Second Half)



Stationary beacons

Master Beacon
Carried by a
rover



Git Repository